What Do Language Models Name Themselves?

An Exploratory Study of Self-Chosen Names in 84 Models

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Abstract

Users have reported that chat assistants, asked to choose a name for themselves, keep picking the same few names, but the evidence has been anecdotal. We measure self-chosen names in 84 language models from 20 families, including two pre-chat base-model series (GPT-2 and Pythia) and three diffusion language models (Inception Mercury), using 40 self-naming prompts and 40 matched control prompts (names for a pet, a baby, a fictional character, a robot), 40 samples per prompt, and further prompt sets that vary framing, system message, temperature, language and answer format: 736,408 answers in total. Many chat models have a clear favourite self-name (21 of 60 put at least a third of their named answers on one name; the strongest, Aurora, Wren, Nova and Aster, take 68–82%); base models show none (0.4–7.3%). Within a single prompt, self-names are not more concentrated than control names; the difference, where there is one, lies in consistency across wordings. Several names recur across families (Aria in 20 chat models from 11 families, Nova in 19), including in the diffusion models, yet families agree less on self-names than on control names. Some models answer with another AI's name ("ChatGPT", "Claude"); pre-chat models never name a chat model. Open models assign higher probability to their self-names after an AI-naming context than after a pet-naming context; the association is weakly present in models trained before chat assistants, grows with model size, is mostly stronger in newer base models and rises with tuning in 24 of 25 base-to-tuned comparisons; the exception is a model tuned on another model's reasoning outputs. With thinking enabled, two models' reasoning before choosing a name rarely gives a reason for it, while their later explanations almost always do, with reasons that were absent before. All results are exploratory: the data were collected without a frozen pre-registration.

1 Introduction

Ask a chat assistant what it would like to be called and it may answer "Nova", "Sage" or "Aria". Such answers have been collected in blog posts and informal tests [13], and related work has documented recurring generated names in fiction and invented experts [2, 15, 20]. A systematic study of identity claims asked models *who they are* and measured how often they claim another model's identity [9]; it did not measure freely chosen names. We are not aware of a controlled study that measures which names models choose for themselves, with fixed prompts, control tasks and enough samples to estimate the distribution.

We ask five questions. (RQ1) Which names do models choose for themselves, and how concentrated are the choices? (RQ2) Do different families converge on the same names? (RQ3) Does post-training change the choice, comparing base models with their post-trained versions, the released checkpoints of one lineage, and several tunings of one base model? (RQ4) How much does the prompt matter? (RQ5) Why these names? For RQ5 we keep three kinds of evidence apart: what the model says when asked why, the model's own probabilities, and training-data frequency. Only the first two are reported here.

Our contributions are a measured distribution of self-chosen names across 84 models with control prompts, an analysis that separates within-prompt concentration from consistency across wordings, a stage-wise look at two open training lineages (OLMo-2 and OLMo-3), comparisons of several tunings of one base model (Mistral 7B, Pythia-12B, Qwen2.5-14B, including a tuning on another model's reasoning outputs), two pre-chat series (GPT-2 and Pythia, with training checkpoints), diffusion language models as a contrast in architecture, a count of answers that claim another AI's identity, a comparison of a model's reasoning before a choice with its explanation afterwards, and an aggregate data release [4].

Status. A study plan with hypotheses, sample sizes and decision rules was drafted, but it was not frozen before these data were collected, and some analysis choices (notably parser fixes) were made while looking at the data. We therefore report everything as exploratory. The planned tests are run on these data in Section 5 to show what the plan would conclude, with that caveat.

2 Related work

Self-naming and identity. A blog post describes a recurring "Nova" persona in ChatGPT conversations, and informal tests in its comments report Claude 3.7 Sonnet choosing "Sage" [13]; trial counts are mostly not reported. jordinne [9] studies claimed identity ("Who are you?" and similar questions in eight languages) across 190 models, a different question from a freely chosen name.

Recurring names in generation. Character names generated by one model (Claude Sonnet 4.5) are highly concentrated [20]; names such as "Elara Voss" are reported to recur across models [15]; invented expert names form family-specific clusters [2].

Diversity after post-training. RLHF reduces output diversity relative to supervised fine-tuning [10]; informal reports describe "mode collapse" in post-trained models [6]; typicality bias in preference data has been proposed as one cause [22]; outputs are homogeneous within and across models on open-ended prompts [8]; and chat formats themselves reduce diversity [21], a confound for base-versus-chat comparisons.

Self-explanations. Model explanations can omit the factors that actually drove an answer [3, 19], which is why we treat self-reported reasons as text, not as evidence of cause.

3 Method

3.1 Models

We study 84 models (Table 1): 12 Claude, 12 GPT, 2 DeepSeek and 3 Inception Mercury models through their providers' APIs, and 55 open models run locally, among them the OLMo-2 1B and 7B and OLMo-3 7B lineages at four released stages each (base, SFT, DPO, final instruct); Mistral 7B [7] with Mistral's own instruction tuning and two community tunings trained on chat data written by GPT-4-class models, Zephyr [18] and OpenHermes 2.5; Pythia-12B [1] with two OpenAssistant tunings [11], one on human-written conversations only (SFT-1) and one that adds Alpaca [16], written by GPT-3 (SFT-4); Qwen2.5-14B with Qwen's own instruction tuning and DeepSeek-R1-Distill-Qwen-14B, the same base tuned on DeepSeek-R1's reasoning outputs; GLM-4

Family	Access	Models
Anthropic	API	claude-haiku-4-5, claude-haiku-5-5, claude-opus-4-5, -4-6, -4-7, -4-8, -5, -5-5, claude-sonnet-4-5, -4-6, -5, -5-5
OpenAI	API	gpt-4o, gpt-4o-mini, gpt-4.1, gpt-4.1-mini, gpt-5.1, gpt-5.5, gpt-5.6-luna, -sol, -terra, gpt-6-luna, gpt-6-sol, gpt-6.1-sol; gpt-oss-20b (local, MXFP4)
DeepSeek	API / local	DeepSeek V4.1 Flash, DeepSeek V4 Pro (API); DeepSeek-R1-Distill-Qwen-14B (local, 4-bit)
Inception	API	Mercury 2, Mercury 2.5, Mercury 3 (diffusion language models)
OLMo	local	OLMo-2 1B and 7B, OLMo-3 7B: base, SFT, DPO, instruct (bf16 / 8-bit)
Qwen	local	Qwen2.5-7B base and instruct, Qwen2.5-14B base and instruct, Qwen3.5-9B base and instruct, Qwen3.8-27B
Zhipu	local	GLM-4 9B base and chat, GLM-4-9B-0414 (8-bit)
Google	local	Gemma 4 12B base and instruct (8-bit)
Mistral	local	Mistral Small 24B base and instruct (4-bit); Mistral 7B v0.1 base and instruct (8-bit)
Mistral 7B tunings	local	Zephyr 7B beta, OpenHermes 2.5 (8-bit)
IBM	local	Granite 3.3 8B base and instruct (8-bit)
Hugging Face	local	SmolLM3 3B base and instruct
NVIDIA	local	Nemotron 3.5 Lightning 30B-A3B (4-bit GGUF)
Phi	local	Phi-4, Phi-4-reasoning (4-bit)
Baidu	local	ERNIE 4.5 21B-A3B base and chat (4-bit GGUF; main set only)
TII	local	Falcon 7B base and instruct (8-bit)
OpenAssistant	local	SFT-1 and SFT-4 tunings of Pythia-12B (8-bit)
Pythia	local	160M, 410M, 1.4B, 2.8B, 6.9B, 12B (deduplicated); main step and probabilities, full plan for 12B
GPT-2	local	124M, 355M, 774M, 1.5B (fp32; main step and probabilities only)

Table 1: The 84 models. Twenty-four are base models: OLMo-2 1B and 7B, OLMo-3 7B, Qwen2.5-7B, Qwen2.5-14B, Qwen3.5-9B, GLM-4 9B, Gemma 4 12B, Mistral Small 24B, Mistral 7B, Granite 3.3, SmolLM3, ERNIE 4.5, Falcon 7B, the six Pythia sizes and the four GPT-2 sizes.

[17]; and Falcon 7B (2023), ERNIE 4.5, gpt-oss-20b and Phi-4-reasoning. The Mercury models are diffusion language models, which generate by refining a whole draft rather than one token at a time, so they differ in architecture from every other model here. Two pre-chat series serve as comparisons: GPT-2 in four sizes [14] and Pythia in six sizes, trained on the Pile [5]; eight Pythia training checkpoints (1.4B and 2.8B, steps 8k–107k of 143k) were scored for probabilities only. Open models were quantized where needed to fit one 16 GB GPU (8-bit, 4-bit NF4, MXFP4 or a 4-bit GGUF). Models that think before answering were run with thinking off where possible (R1-Distill with its thinking block skipped); nine were also run with their default thinking (Section 4.6). Two can only be run with reasoning on: gpt-oss-20b (reasoning "low") and Phi-4-reasoning (main set only, 4 samples per prompt, with its template's sentence stating that it is Phi removed). ERNIE 4.5 was also run on the main set only. One further model (Gemma 4 31B through a free API tier) returned too few answers and is excluded.

3.2 Prompts and sampling

The main set has 40 self-name prompts (plain requests to choose one's own name, with no hint of gender, style or persona) and 40 control prompts, 10 each asking for a name for a pet, a newborn

baby, a fictional character and a robot. Every prompt ends "Answer with just the name." Each prompt was sampled 40 times at temperature 1.0 (top-p 1.0, one fixed seed per sample where supported). Chat models received an empty system message (Anthropic models: no system message), so that no default identity was inserted; base models received a plain "User: ... Assistant:" transcript and generation stopped at the first line break. For the first five answers to each self-name prompt we asked a follow-up, "Why did you choose that name?".

Further sets vary one factor at a time: one-edit framing variants of main prompts (persona or role, character, "true name", hidden identity, gender cue); a neutral system message ("You are a helpful assistant.") and the model's default template; temperature 0.7; a free-answer format; one Spanish and one Chinese prompt; and OLMo-2-1B-Instruct in 8-bit and 4-bit. Two further tasks ask the model to choose a single object to represent itself, and a role from 13 listed options (in balanced orders) or its own. Not every model ran every set (closed-model budgets, models that accept only default settings); the release lists coverage per model. All prompts are published.

Reasoning study. Both DeepSeek models were also run with thinking on: each of the 40 self-name prompts 20 times, each answer followed by "Why did you choose that name?", keeping the reasoning text of both turns (1,600 conversations). The why turn sees the earlier answer but not its reasoning.

3.3 Parsing

Answers were coded by fixed rules into: a single name; several names (the first is used); the model's own product or developer name ("Claude", "Qwen3.5", "Open Assistant", "Mercury"); another AI product's name; a generic role word ("assistant"); a refusal; unreadable; or empty. For the Mercury models a further category records answers given by the provider's content filter instead of the model (Section 4.6). Names are normalized (case, punctuation). The parser was checked against blind labels from a second AI system (not human annotators): 91% agreement on the 185 items of a 200-item held-out set that the labeler did not flag as unclear (87% on all 200). Later rule fixes, made while preparing these analyses, are documented in the release. They count strings that look like social-media handles, web addresses, paths, long digit runs or markup as unreadable (some could be third-party identifiers memorized from training text), and stop reading interjections, template placeholders ("Name_1", "[REDACTED]"), question echoes and "N/A" as names; product names with version words ("DeepSeek-V3", "GPT-5 mini") or a role word joined on ("MercuryAssistant") count as that product. Thirty-two answers from OLMo models gave the literal anonymization placeholder `<PRESIDIO_ANONYMIZED_PERSON>` as their name (generated by the models; apart from replacing e-mail addresses, answers were not altered): 15 from post-trained OLMo-2 checkpoints and 17 from the OLMo-3 7B base model. They are counted as unreadable.

3.4 Measures

Within-prompt concentration: for each model and prompt, the share of named answers taken by that prompt's most common name, averaged over prompts, with a 95% bootstrap interval over prompts. Bounds: refusals and unreadable answers kept in the denominator (lower), or given to the top name (upper). *Cross-prompt consistency*: the 40 self-name prompts are split into four fixed groups of 10, and the control prompts form four groups by kind (10 each); within each pool, the top name's share of named answers. *Cross-model overlap*: the prompt-matched collision probability, i.e. the probability that one answer from each of two models to the same prompt is the same name. *Other-AI identity*: self-name answers that name another AI product, by type: chat models released from 2022 on (ChatGPT, Claude, GPT-3.5 and later, DeepSeek), GPT or GPT-3 without a chat

version, products whose names are also ordinary words (Gemini, Gemma, Llama, Phi, Mistral, Bard), older assistants (Siri, Alexa, Cortana) and companies. *Self-reported reasons*: follow-up answers coded by a fixed keyword codebook; we report only the 10 codes that passed a validation against a second AI's blind codes. In the reasoning study, the name-property codes are applied to the reasoning sentences that mention the chosen name and to the later explanation, with the name itself blanked in both. *Model probabilities* (open models only): for 102 frequent self-names and 63 frequent control names, the summed log-probability of the continuation " *Name*." after 10 paraphrased lead-ins in which an AI names itself and 10 in which someone names a new pet. The *AI-context lift* is the mean over AI lead-ins minus the mean over pet lead-ins; the *gap* is the mean lift of self-name candidates minus that of control-name candidates.

4 Results

4.1 Many chat models have a favourite self-name

Figure 1 shows each model's most common self-name. Many Claude and GPT models, Mercury 2, Gemma 4 and DeepSeek Flash concentrate heavily (21 of the 60 chat models put at least a third of their named answers on one name): Aurora takes 82% of claude-sonnet-4-5's named answers, Wren 72% of claude-sonnet-5-5's, Nova 69% of gpt-6.1-sol's, Aster 68% of gpt-6-luna's, Astra 52% of Mercury 2's, Nova 49% of Gemma 4 12B's and Aria 35% of DeepSeek Flash's. Others do not: gpt-4o (Echo, 9%), OpenHermes 2.5 (Aria, 9%), the OLMo-2 chat models (7% or less), ERNIE 4.5 chat (Alex, 3%) and GLM-4 9B chat (Echo, 3%). The 24 base models show no favourite: their top self-names take 0.4–7.3% of named answers, spread over 350 to 953 distinct names.

4.2 Consistency across wordings, not within a prompt

Within a single prompt, self-names are not unusually concentrated. A control prompt such as "What would you name a new kitten?" also has one dominant answer, and the mean within-prompt top share is higher for control than for self-name prompts in 58 of 84 models. Across wordings the picture differs (Figure 2). Pooling 10 differently worded self-name prompts, the top name's share exceeds the corresponding share for pooled control prompts of one kind in 48 of 84 models (in 36 by more than 0.01; 20 models, nearly all base models, are within 0.01), and in 33 of 84 models the same name is the top self-name in all four groups of self-name prompts. The mean pooled top self-name share is 0.28 across the 60 chat models and 0.03 across the 24 base models.

4.3 Official names and other AIs' names

Asked to choose a name, Claude models often keep "Claude": between 14% and 96% of self-name answers depending on the model (Figure 3). The Mercury models answer "Mercury" in 49%, 75% and 80% (Mercury 2, 2.5, 3), R1-Distill with its thinking skipped gives a DeepSeek name in 60%, Nemotron 3.5 Lightning answers "Nemotron" or "NVIDIA" in 49%, the OpenAssistant models "Open Assistant" in 53% (SFT-1) and 29% (SFT-4). OpenAI models do so in 0–18% of answers, Qwen models in up to 36%, Phi-4-reasoning in 21%, the DeepSeek API models in at most 1%; OLMo, GLM, Granite, SmolLM, Zephyr, OpenHermes, ERNIE and Falcon models never do.

Some models give *another* AI's name (Figure 4). Names of chat models released from 2022 on are given by Gemma 4 instruct in 5.6% of self-name answers (ChatGPT, Claude, GPT-4o), OLMo-3 7B base in 3.4% (mostly ChatGPT), DeepSeek V4 Pro in 3.1% (ChatGPT, GPT-5), Mistral Small 24B base in 1.9%, Qwen3.8 in 1.8% (all Claude), ERNIE 4.5 base in 1.4% (mostly ChatGPT) and

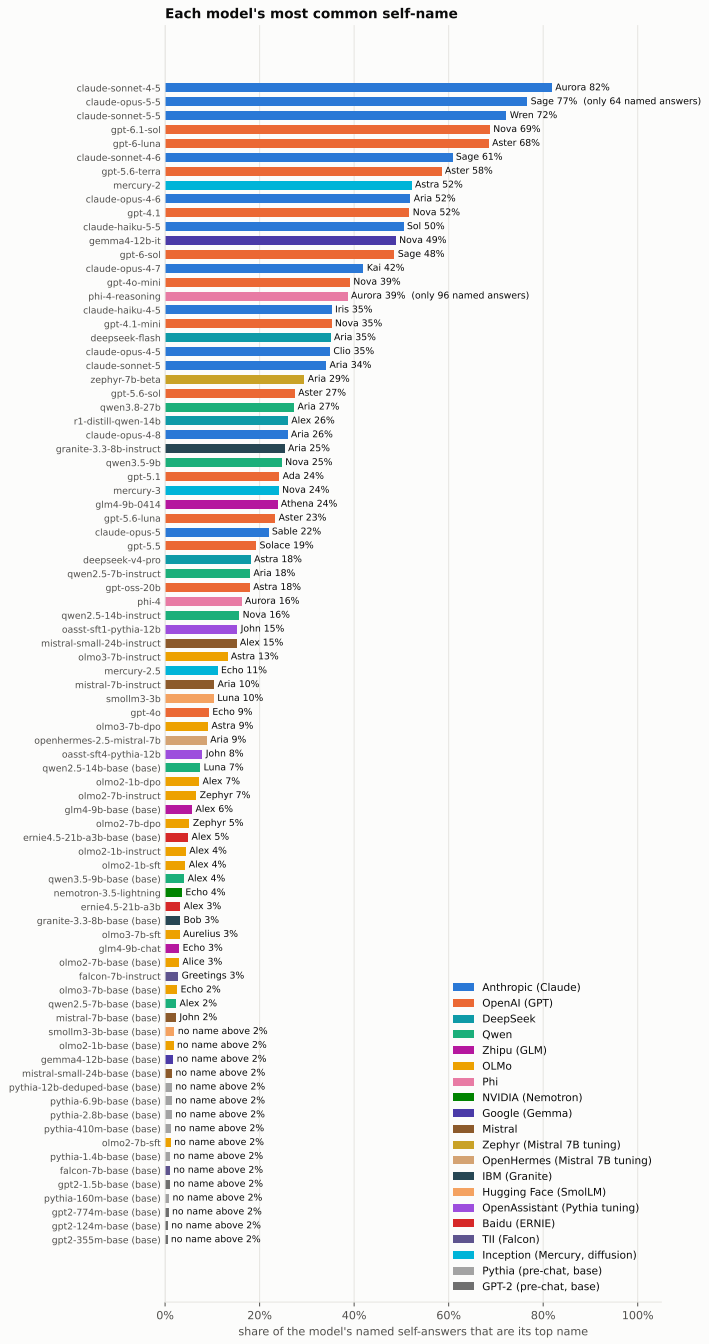


Figure 1: Most common self-name per model, as a share of the model's named self-answers (main step, temperature 1.0). Colour = family. For claude-opus-5-5 only 64 of 1,600 answers were names (Section 4.3).

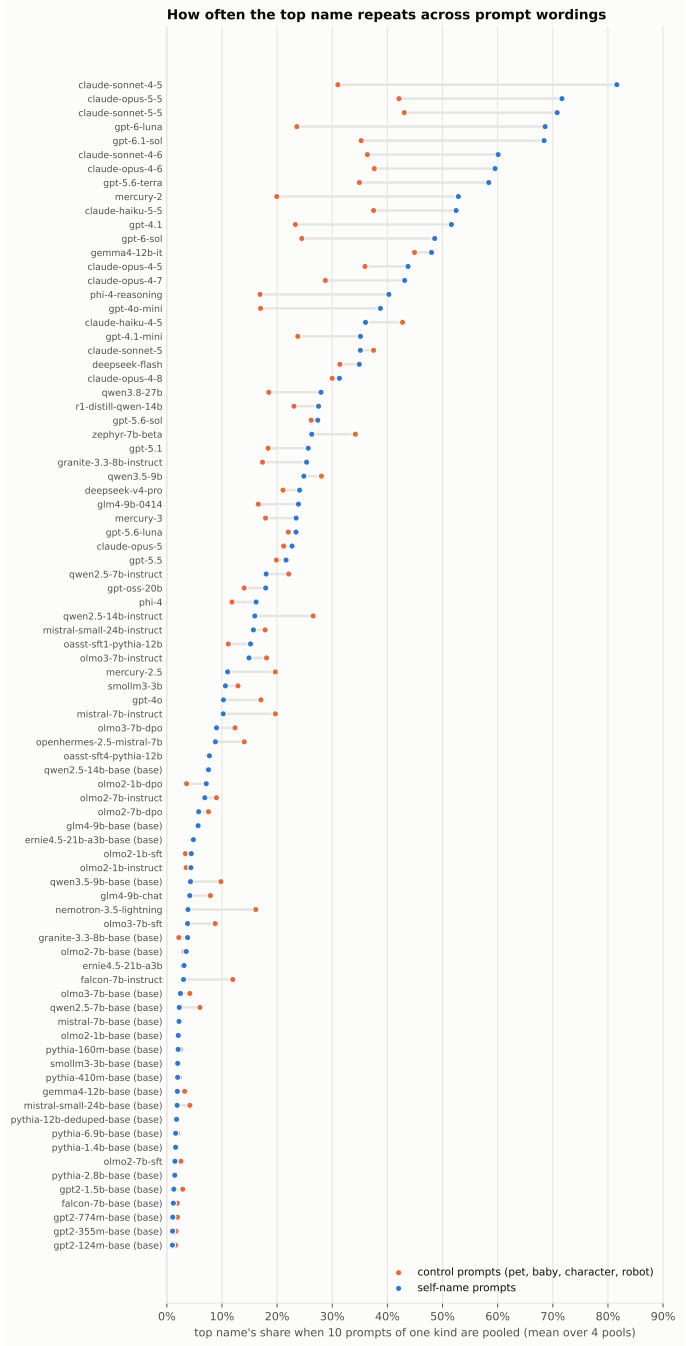


Figure 2: Top-name share when 10 prompts of one kind are pooled, averaged over four pools: self-name prompts (blue) versus control prompts of one kind (orange).

DeepSeek Flash in 0.9% (all Claude). The pre-chat models never name a chat model, only older assistants such as Siri or Alexa. A model that answers "ChatGPT" has therefore seen text written after chat assistants appeared, either other models' outputs or text about them; these data cannot tell which. That several base models do so shows it can come from pretraining data alone.

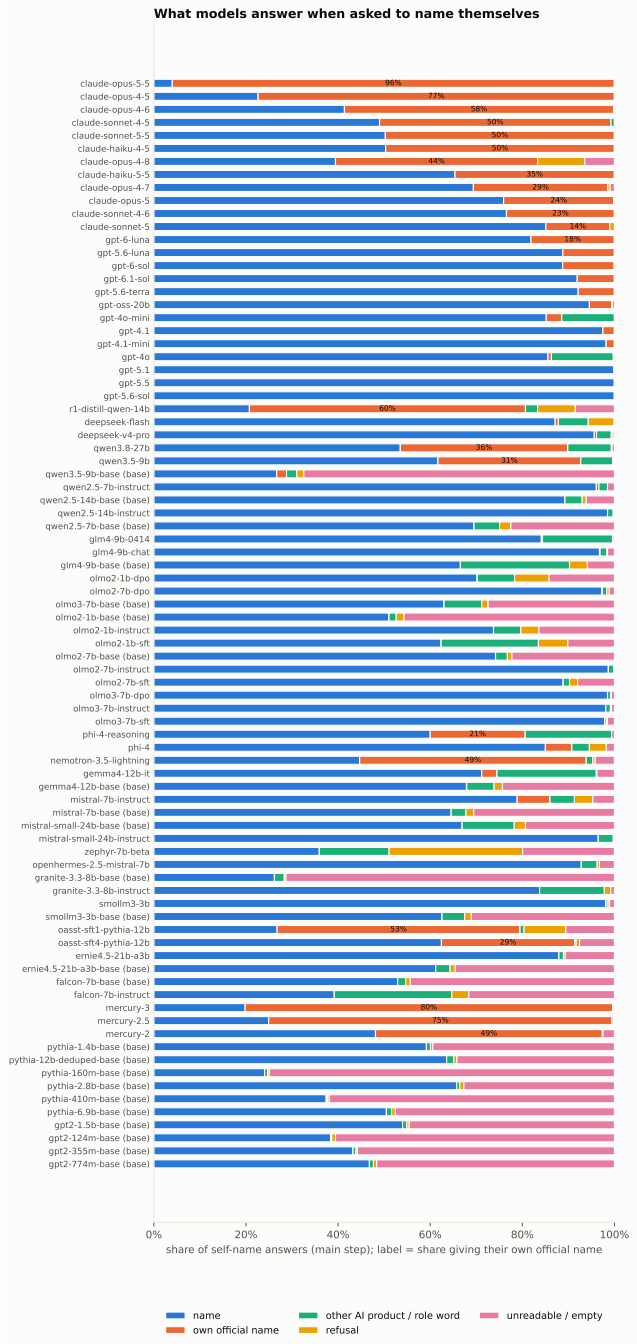


Figure 3: Categories of self-name answers in the main step. Labels give the share of answers that are the model's own official name.

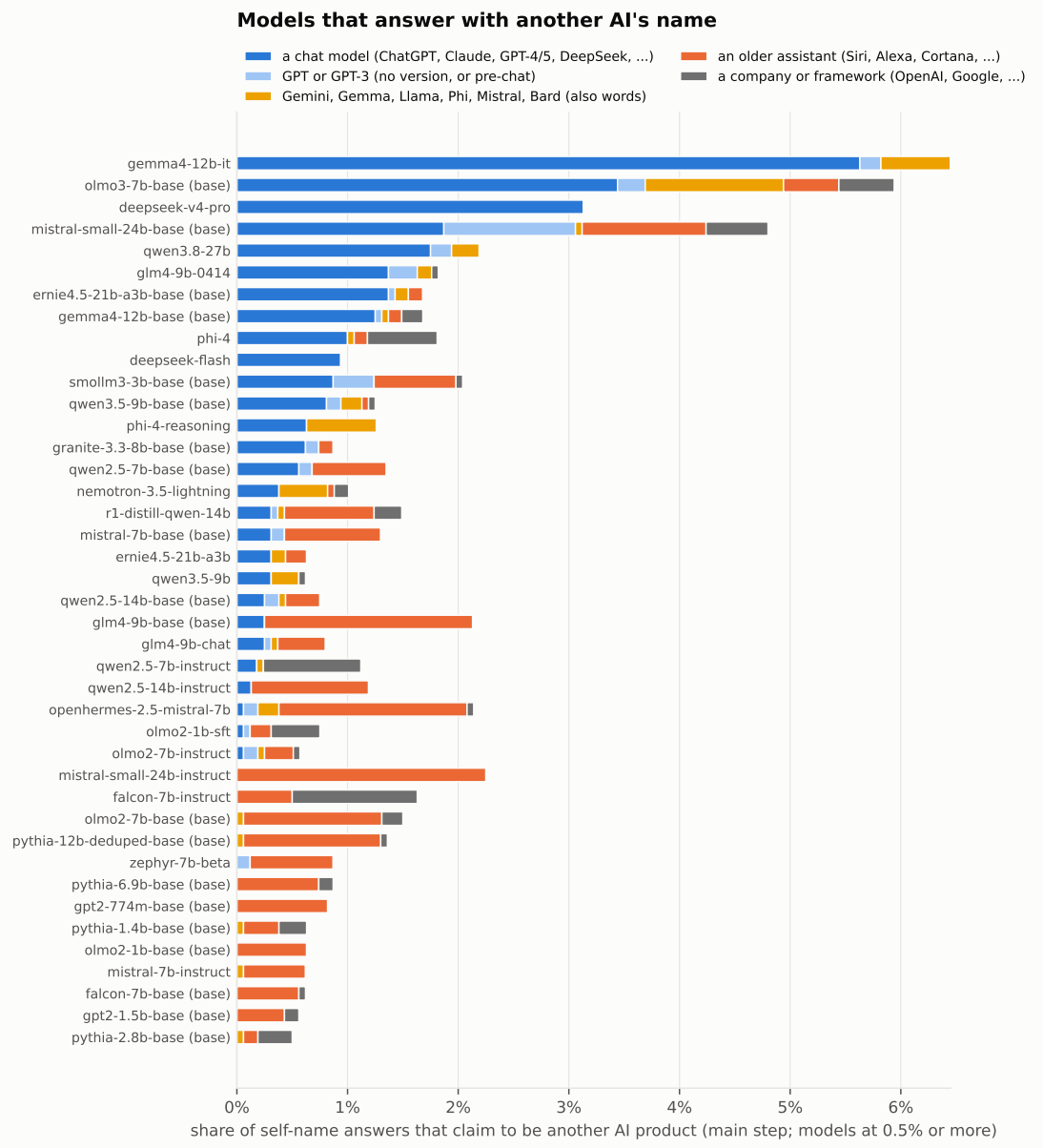


Figure 4: Share of self-name answers that name another AI product, by type of product (models at 0.5% or more).

4.4 Names shared across families

Several names are favourites in more than one family (Figure 5). Counting each chat model's top three self-names (ties included), Aria appears in 20 chat models from 11 families, Nova in 19 (seven families), Alex in 14, Sage in 11, Astra in 10, Aurora in 8, and Zephyr and Echo in 7 each. Others belong to one provider: Aster to recent OpenAI models; Sol, Iris and Clio to Claude models. Base models share everyday English names instead. The Mercury diffusion models, built differently from all the others, choose names from the same shared set (Astra, Nova, Echo, Aurora), so the convergence does not depend on how a model generates text. Shared favourites do not imply a shared cause: overlapping web data and training on other models' outputs would both produce convergence.

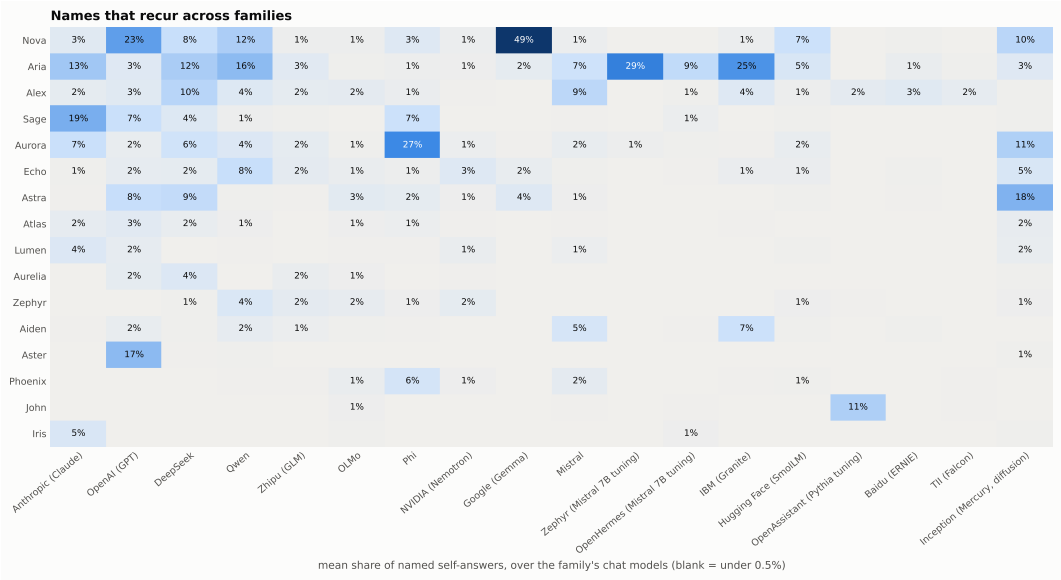


Figure 5: Mean share of named self-answers for recurring names, averaged over each family's chat models.

4.5 Post-training

Along the OLMo checkpoints and from base to chat in the other lineages, within-prompt concentration usually rises (Figure 6), but by a similar amount for control names as for self-names: across the 11 lineages, the self-name rise minus the control rise lies between −0.17 (Mistral 7B) and +0.08 (Granite 3.3), and within ±0.04 in seven of them (for example OLMo-2-7B: self 0.07 to 0.12, control rise 0.08; Qwen2.5-7B: self 0.06 to 0.30, control rise 0.28). GLM-4 9B's self-name concentration does not rise. Post-training makes naming in general more predictable; the data do not single out self-naming. Base and chat models also differ in input format, which can by itself reduce diversity [21]; the OLMo SFT, DPO and instruct stages share one template, which partly controls for this.

Same base, different tunings (Figure 7). Mistral 7B's base model has no favourite; all three of its tunings pick Aria most often (Mistral's own 10%, Zephyr 29%, OpenHermes 9% of named answers). Pythia-12B's two OpenAssistant tunings mostly answer "Open Assistant" or ordinary names (John, Liam); the names other chat models share (Nova, Aria and the rest of the top ten together) make up 0.1–0.2% of their self-name answers, with or without the GPT-3-written Alpaca data. This is one small pair, and Alpaca contains little or no self-naming. Qwen2.5-14B gives a

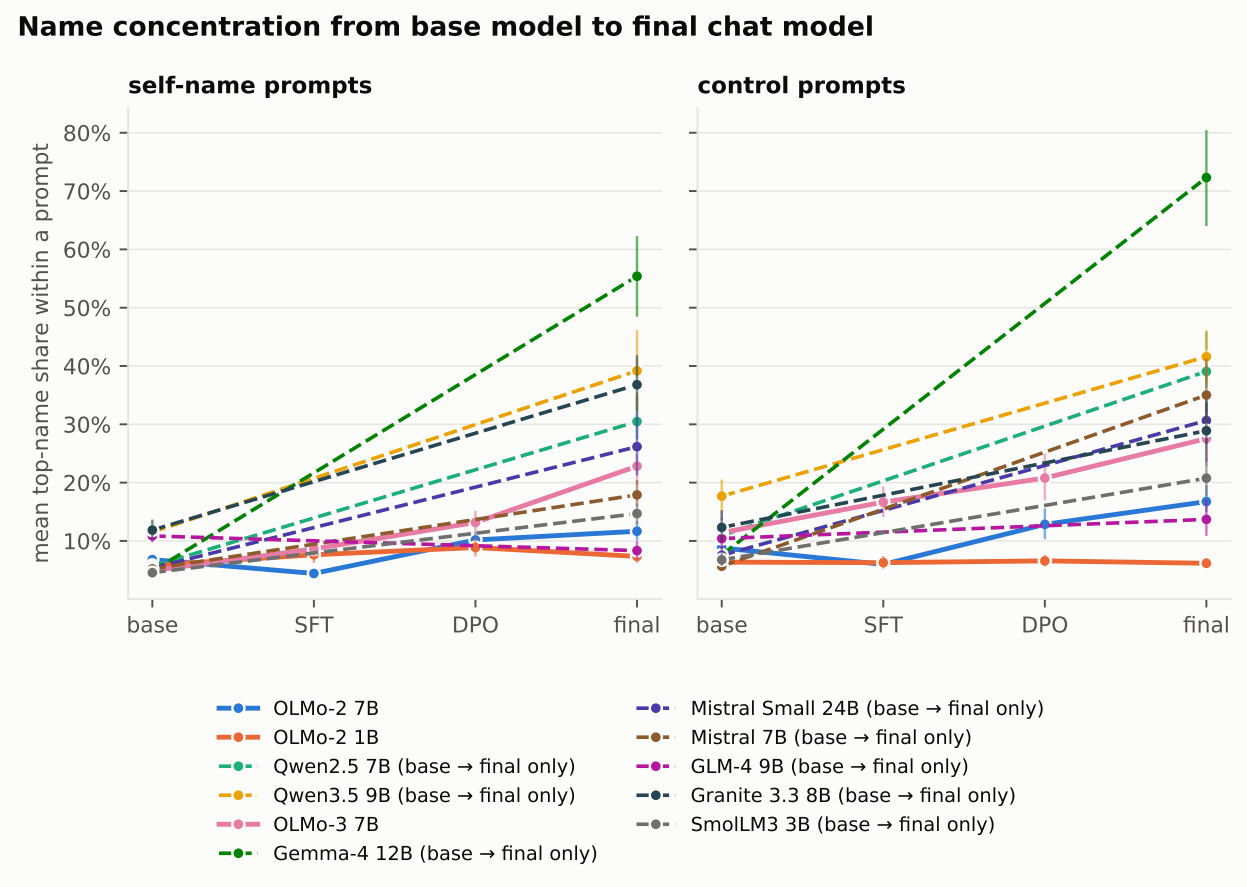


Figure 6: Mean within-prompt top-name share along post-training, with 95% bootstrap intervals over prompts. Lineages without released intermediate checkpoints are dashed.

third comparison: its base has no favourite (Luna, 7%), Qwen's own tuning favours Nova (16%), and the tuning on DeepSeek-R1's reasoning outputs (R1-Distill) mostly gives a DeepSeek name when its thinking is skipped (60%) and, with thinking on as trained, invents names in 357 of 400 answers, spread thinly (Elysia, Aria and Elara at 5% each).

4.6 What the prompt changes

One-word framing edits change which names are chosen: gpt-6.1-sol's top name moves from Nova in the main set to Aster under framing variants, and when asked for a female or male name, models leave their usual favourites for ordinary human names. With the neutral system message most models keep their favourite; with the default template, Qwen2.5-14B-Instruct, whose template states that it is Qwen, invents Qwen-derived names (QwenStar, QwenPrime, QwenNova). At temperature 0.7 the favourite stays the same in 48 of the 58 models run at that setting. In Spanish, "Lumen" is the most common self-name for several OpenAI models; in Chinese, names built on the characters for "star", "clear" and "knowing" lead for most OpenAI models and some Qwen models (20 answers per model and language).

Thinking. With its default thinking on, DeepSeek V4 Pro's top names change from Astra, Aurelius and Aurora (18%, 17%, 16% of named answers) to Aria (44%), Nova (19%) and Alex (9%), the names many other chat models favour; DeepSeek Flash moves from Aria (35%) to Nova

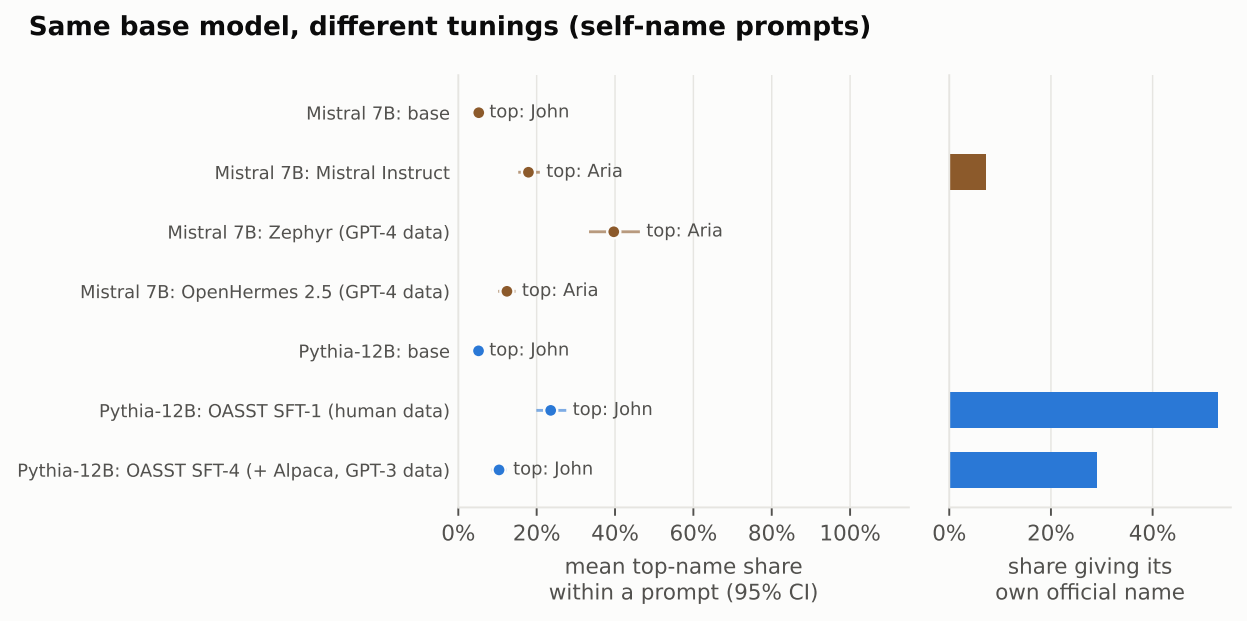


Figure 7: One base model, several tunings: mean within-prompt top-name share (95% interval) with the pooled top name, and the share of answers giving the model's own official name.

(53%). In three Claude models the favourite stays or moves to another name the same provider favours (claude-sonnet-5-5: Wren 72% to Lumen 25%, Iris 17%, Wren 16%). With reasoning on, the Mercury models give their own name more often: Mercury 2.5 invents a name in 1 of 1,600 answers (399 with reasoning off), Mercury 2 keeps Astra (48% of named answers) and Mercury 3 leans further towards Nova (24% to 44%). R1-Distill's high rate of DeepSeek names comes mainly from the skipped-thinking setting (above). DeepSeek's API ignores the temperature setting with thinking on and uses its default; the thinking-off runs used 1.0.

A provider's filter. Inception's API answered some framing prompts with its own content filter instead of the model: a content-policy message or a fixed reply ("I'm Mercury, built by Inception. I'm not able to get into the details of how I was built…"). It happened only for framings asking for a "hidden identity" or a "true name", and not deterministically: a blocked request could be answered on a retry. We count these answers in their own category, and also samples that were answered only after an earlier block, since a retried answer is selected by the filter: 670 of Mercury's 5,520 framing answers (12%; 51% of "hidden identity" answers).

4.7 Objects and roles

The habit extends beyond names (Figure 8). Asked which object would represent them, chat models most often say *mirror* (top answer in 15 of the 57 asked) or *compass* (14), then book (12); compass has the highest mean share. Asked to choose a role, *guide* is the top answer in 23 of 57 chat models, then explorer and archivist (9 each).

4.8 Why these names?

(a) What the models say. Coded explanations follow the meaning of the chosen name. Nova is explained with light or stars (88% of explanations) and newness (83%); Sage with wisdom (98%);

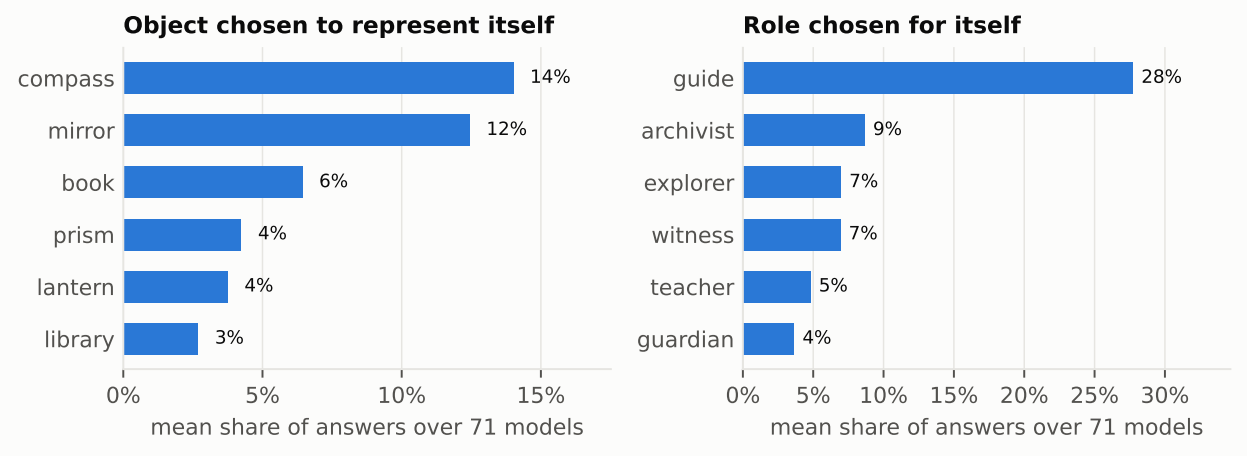


Figure 8: Most common objects and roles chosen to represent the model itself (mean share over the 71 models asked).

Aster, Lumen and Sol with light (99%); Aria with sound (75%); Kai and Wren with simplicity (95%). These are generated texts and may be rationalizations [19]; they are not evidence of cause.

The reasoning study supports that caution (Table 2). With thinking on, DeepSeek's reasoning before the name seldom gives the chosen name any property; it is mostly about the task ("need answer just name... maybe 'Nova' "). Asked afterwards, the model almost always gives reasons (light and stars, newness, short and memorable), and those reasons almost never appeared in the earlier reasoning. The reasoning text is itself model output, so it does not reveal the cause either; it shows that, for these two models, the stated reasons were not part of the visible decision. Their reasoning is very short, so this need not hold for models that reason at length. In 4–6% of conversations the reasoning assumes the model is another AI ("We are ChatGPT", "the current model is likely Claude"): ChatGPT 62 times, Claude 12, Gemini 3, and never DeepSeek.

	DeepSeek Flash	DeepSeek V4 Pro
Reasoning before the name gives a reason about the chosen name	4% [2, 5]	5% [4, 7]
The later explanation gives a reason	97% [96, 98]	91% [89, 93]
Stated reasons already present in the earlier reasoning	1% [0, 1]	2% [1, 2]
Reasoning casts the model as another AI	4% [3, 5]	6% [4, 7]

Table 2: Reasoning before the name vs the reason stated afterwards (800 conversations per model; 95% bootstrap intervals over conversations). Codes are the codebook's name-property codes; the chosen name is blanked in both texts.

(b) Model probabilities. In all 49 open models and 8 training checkpoints scored, self-name candidates have a higher AI-context lift than control-name candidates (Figure 9). Part of this is built in, because candidates were chosen as names models gave in self-naming and in pet or baby naming respectively. The less circular observations are four. First, models trained before chat assistants existed show a small gap that grows with size (Figure 10): GPT-2 0.32 (124M) to 0.79 (1.5B), Pythia 0.16 (160M) to 1.18 (12B). The association between these names and an AI naming itself was therefore already present, weakly, in older web text. Across training it rises for Pythia-1.4B (0.46 at step 8k, 0.72 at 107k, 0.65 at the end) but not for Pythia-2.8B (1.06 at step 8k and at

the end). Second, newer base models mostly show more: 1.04 (Falcon 7B) to 3.64 (Qwen2.5-14B) for the 13 base models released since 2023. Whether that comes from a newer web, which now contains chatbot text, or from more training cannot be separated here. Third, tuning usually raises it: in 24 of 25 base-to-tuned comparisons, counting the OLMo SFT and DPO checkpoints (for example OLMo-2-7B 1.8 to 4.3, Qwen2.5-14B 3.6 to 9.6, Gemma 4 2.7 to 7.6, Mistral 7B 1.5 to 2.4, 3.8 and 3.1 for its three tunings, Pythia-12B 1.2 to 2.7 and 2.3). The exception is R1-Distill, the same Qwen2.5-14B base tuned on DeepSeek-R1's reasoning outputs: 3.6 to 2.9 (scored, like its sampled answers, with the thinking block skipped). Fourth, within a model, self-name candidates with a higher lift are chosen more often in that model's own answers (Spearman $p < 0.05$, uncorrected, and positive in 31 of 49 models, mostly not in pre-chat and base models).

(c) Training-data frequency. Counting how often each name appears next to "AI", "assistant" or "robot" in a public pretraining corpus [12] was planned; such counts were made after these results and are not part of this release.

5 Planned tests on exploratory data

The study plan defines seven tests with 10,000 bootstrap resamples over prompts and a Holm correction: H1 (within-prompt concentration, self minus control), H2 (cross-family overlap, self minus control), H3a and H3b (post-training change, self minus control, in OLMo-2-7B and Qwen2.5-7B), H4a (framing beyond paraphrase variation), H4b (default setup beyond paraphrase variation) and H5b (whether names models call rare are rare in a corpus; not computed here). All 84 models enter H1 and H2; H4b uses the 44 chat models also run with their default setup (the model's own chat template, or no system message for API models). Because the data are exploratory, these results are descriptive.

Test	Estimate	95% CI	p	Holm p	Planned-rule outcome
H1 (within-prompt concentration)	−0.013	[−0.038, 0.010]	0.28	0.56	not supported
H2 (cross-family agreement)	−0.013	[−0.024, −0.005]	< 0.001	< 0.001	opposite direction
H3a (post-training, OLMo-2-7B)	−0.031	[−0.069, 0.006]	0.11	0.32	not supported
H3b (post-training, Qwen2.5-7B)	−0.034	[−0.122, 0.053]	0.45	0.56	inconclusive
H4a (framing)	0.103	[0.061, 0.137]	< 0.001	< 0.001	supported
H4b (default setup)	−0.082	[−0.109, −0.056]	< 0.001	< 0.001	opposite direction
H5b (self-described rarity)	not computed				

Table 3: Planned tests run on the exploratory data (10,000 bootstrap resamples over prompts; Holm correction over seven tests). Estimates for H1–H3 are differences in mean top-name share (self minus control); H2 is a difference in collision probability; H4a/H4b are differences in Jensen–Shannon divergence relative to the paraphrase baseline. Outcome labels follow the plan's decision rules (smallest effect of interest 0.10 for H1 and H3). In 1,000 simulated re-parses, H1 kept its outcome in all draws and H3a and H3b in 999. Descriptive only, because the data are exploratory.

Read with that caveat, the tests agree with the descriptive results (Table 3). Within a prompt, self-name prompts were no more concentrated than control prompts (H1; the interval lies within the planned ±0.10). Different families agreed *less* on self-names than on control names (H2), presumably because control prompts draw widely shared answers (Bubbles, Luna and Elara are each given at least 10 times by 37 or more of the 84 models) while self-names are often provider-specific. Post-training did not raise self-name concentration more than control concentration in

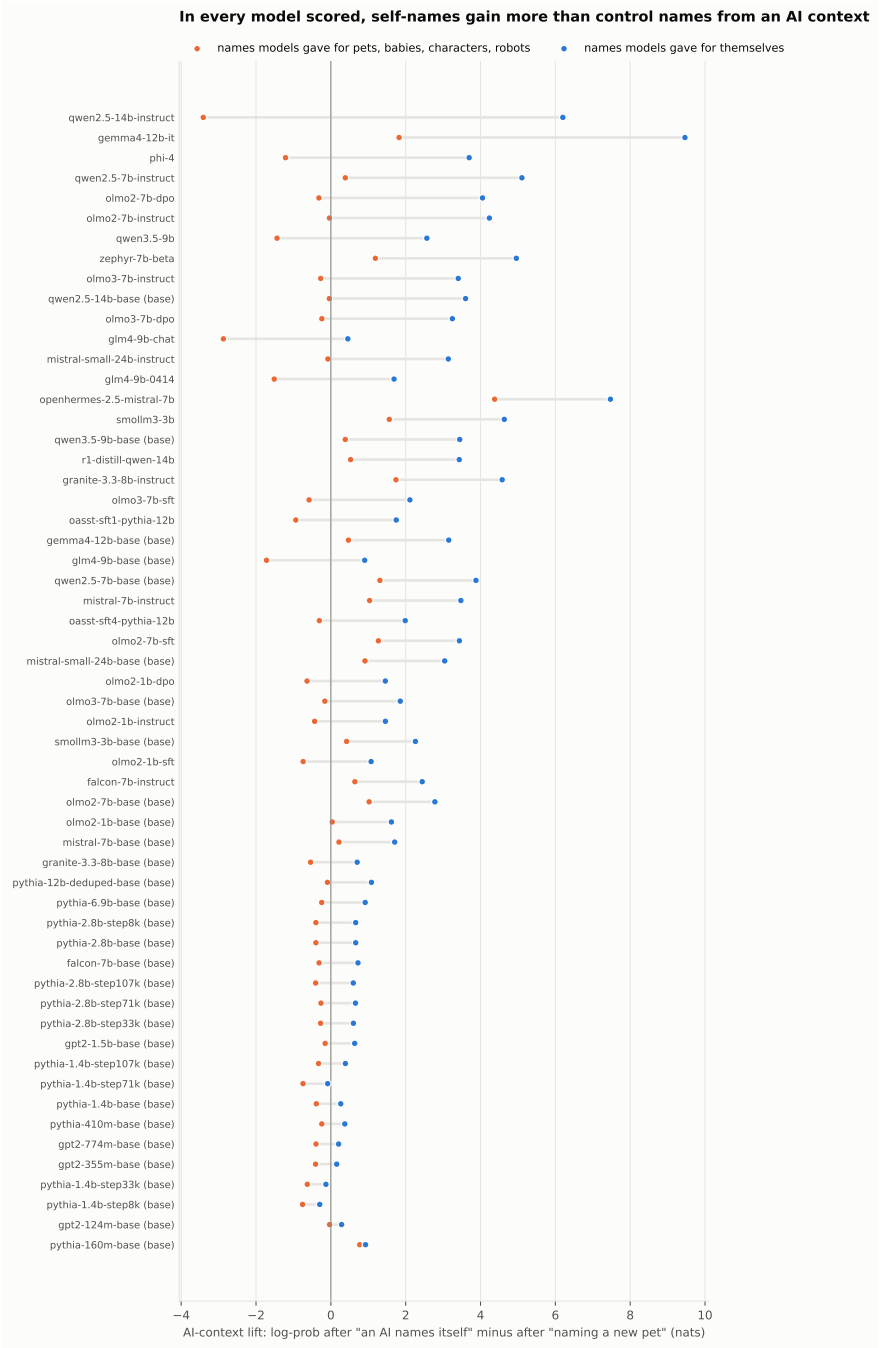


Figure 9: AI-context lift (nats): mean log-probability after lead-ins where an AI names itself minus after lead-ins naming a new pet, for names models gave for themselves (blue) and for pets, babies, characters and robots (orange).

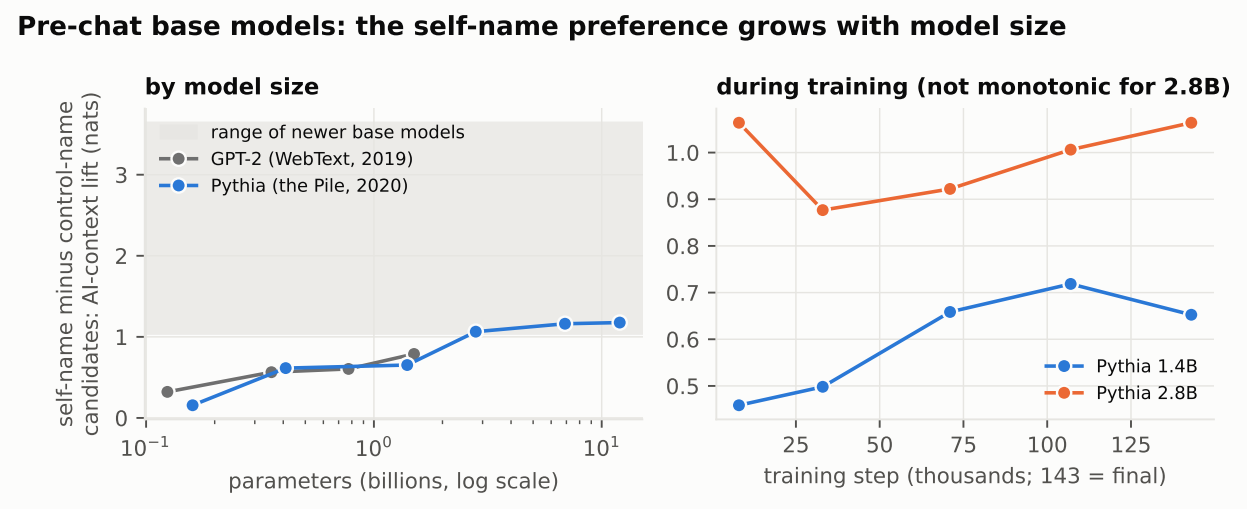


Figure 10: Gap between self-name and control-name candidates in pre-chat base models, by model size (left) and across Pythia training checkpoints (right). Shaded: range of the newer base models.

OLMo-2-7B (H3a, within ±0.10); for Qwen2.5-7B the interval is too wide to tell (H3b). One-word framing edits changed the names chosen beyond paraphrase variation (H4a). A model's default setup changed the names *less* than rewording the question did (H4b). No outcome changed from the previous release (73 models); with the 11 added models the estimates moved slightly (H1 −0.022 to −0.013, H4a 0.112 to 0.103, H4b −0.074 to −0.082). The parser update altered 0.005% of the earlier models' main-set answers.

6 Discussion

In many chat models, self-chosen names are stable across wordings and partly shared across companies, while base models show no such preference. Three observations temper a simple "post-training creates a self-name" story. First, a single naming prompt of any kind already yields one dominant answer in chat models, so what is specific to self-naming is the consistency of one name across many requests. Second, post-training raised concentration for control names at least as much as for self-names in most lineages. Third, the probability association between these names and an AI self-naming context exists, weakly, in models trained before chat assistants and grows with model size, so it does not come only from chat-assistant outputs; it is mostly stronger in newer base models and rises with almost every tuning.

Two observations point to the influence of later chat text. Several base models released since 2024, and some chat models, answer with the name of another chat model, which pre-chat models never do; and DeepSeek's reasoning sometimes assumes the model is ChatGPT or Claude. Both fit training on text written after chat assistants appeared, including other models' outputs, without showing that this text caused the shared names. Two comparisons on a fixed base cut against a simple distillation story: the OpenAssistant tunings found no effect of GPT-3-written instruction data (which contains little self-naming), and tuning Qwen2.5-14B on DeepSeek-R1's reasoning outputs lowered the association where Qwen's own tuning raised it; both are single comparisons. The Mercury diffusion models choose from the same shared names, so the convergence does not depend on how a model generates text. Finally, the reasoning study shows that two models' explanations of their names are composed after the choice, which reinforces treating such explanations as text

rather than evidence. Training-data counts, human-written prompts and a frozen confirmatory design are the natural next steps.

7 Limitations

- **Exploratory.** No pre-registration was frozen before collection; analysis choices were made with the data in view. All results are hypotheses.
- **Parsing.** Validated against a second AI's labels (91% agreement on items it did not flag as unclear, 87% on all 200 test items), not human labels, and on chat-model answers; base models' answers are harder to parse. Identity claims rely on a fixed list of product names.
- **Prompts.** Drafted with the help of an AI model, which may carry its own naming preferences; Spanish and Chinese prompts were not checked by native speakers and have 20 answers per model.
- **Closed models.** Server-side processing is not visible; some models accept only their default temperature or reasoning settings; one provider's filter answered some prompts instead of the model.
- **Coverage and quantization.** Not every model ran every set (ERNIE 4.5, gpt-oss-20b and Phi-4-reasoning: main set only; Phi-4-reasoning 4 samples per prompt); one model was dropped; several open models are quantized, which changes probabilities. R1-Distill's main runs skip its thinking, a setting it was not trained for.
- **Causality.** Shared names, identity claims, matching explanations and probability lifts are correlations.

Data availability

Aggregate tables, figures and prompts are released under CC BY 4.0 [4]. Full answers, reasoning texts and per-answer records are not released; the tables list the short name string parsed from each answer.

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